

Métodos Matemáticos de la Física II

Part I: Structure of Banach and Hilbert spaces

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(Dated: May 19, 2026)

Bla bla bla... resumen...

I. METRIC SPACE

A set X is called a *metric space* if it is equipped with a notion of *metric distance*, i.e. a function $d : X \times X \rightarrow \mathbb{R}$ such that, for any three elements x, y and z in X , which are called *points*, it holds:

1. $d(x, y) \geq 0$, i.e. it is *non-negative*.
2. $d(x, y) = 0$ implies $x = y$, i.e. it is *positive definite*.
3. $d(x, y) = d(y, x)$, i.e. it is *symmetric*.
4. $d(x, z) \leq d(x, y) + d(y, z)$, i.e. it satisfies the *triangle inequality*.

Example 1: The set $\mathbb{R}^n$ with the euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ is a metric space.

Example 2: For any $p \geq 1$, the set

$$L^p(\mathbb{R}^n) = \left\{ f \in C(\mathbb{R}^n) : \int_{\mathbb{R}^n} dx |f(x)|^p < \infty \right\}$$

is a metric space with metric distance

$$d(f, g) = \left(\int_{\mathbb{R}^n} dx |f(x) - g(x)|^p \right)^{1/p}.$$

At difference with previous example, this is a metric space that is associated to an infinite-dimensional vector space.

A. Convergence

We say that a sequence $\{x_j \in X : j \in \mathbb{N}\}$ *converges* to a value $x \in X$ if, for each $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$. It can be easily shown that, when the sequence is convergent, then the value of x is unique.

The convergence of the sequence is abbreviated by $x_j \rightarrow x$ as $j \rightarrow \infty$, or even more simply by just $x_j \rightarrow x$.

B. Continuous function

The notion of convergence enables the definition of the continuity of a function. Namely, consider two metric spaces X and Y . A function $f : X \rightarrow Y$ is continuous at $x \in X$ if, for every convergent sequence $x_j \rightarrow x$ it holds that $f(x_j) \rightarrow f(x)$. Although continuity will not be used in the remainder of Part I, it is introduced here because it motivates the definition of convergence.

C. Cauchy sequence

A sequence $\{x_j \in X : j \in \mathbb{N}\}$ is a *Cauchy sequence* if, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_i, x_j) < \epsilon$ for every $i, j \geq N$.

Excercise: Show that a convergent sequence is a Cauchy sequence.

Remark: Not every Cauchy sequence in a metric space X converges to a value in X . The following example illustrates this idea.

Example: Consider the sequence given by

$$x_j = \sum_{i=0}^j \frac{1}{i!}.$$

Clearly, $x_j \rightarrow e$ as $j \rightarrow \infty$, i.e. it converges to Euler's number $e = 2.71828\dots$. This is a Cauchy sequence in $\mathbb{Q}$ but it does not converge to a value in $\mathbb{Q}$ because e is irrational. In other words, it converges to a value in $\mathbb{R}$ that is not in $\mathbb{Q}$.

D. Complete metric space

A metric space X equipped with a metric d is called *complete* if every Cauchy sequence in it converges to an element in it. More formally, if X is a metric space, then for any Cauchy sequence $\{x_j \in X : j \in \mathbb{N}\}$, there exists an $x \in X$ such that, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$.

Remark: The example of previous section, IC implies that $\mathbb{Q}$ is not a complete space.

II. NORMED VECTOR SPACE

A *normed vector space* V over a field $\mathbb{F}$ is a vector space equipped with a norm, which is a mapping $V \ni v \rightarrow |v| \in \mathbb{R}$ that, for any $v, w \in V$ and $\alpha \in \mathbb{F}$, it satisfies:

1. triangle inequality, $|v + w| \leq |v| + |w|$,
2. absolute homogeneity, $|\alpha v| = |\alpha||v|$, and
3. positive definiteness, $|v| = 0$ implies $v = 0$.

In particular, a *semi-norm* is a norm without property 3.

Example 1: In correspondence with the first example of previous section,

$$|x| = \sqrt{\sum_{i=1}^n x_i^2}$$

defines a norm over $\mathbb{R}^n$.

Example 2: In correspondence with the second example of previous section,

$$|f| = \left(\int_{\mathbb{R}^n} dx |f(x)|^p \right)^{1/p}$$

defines a norm over $L^p(\mathbb{R}^n)$.

A. A normed vector space is a metric space

A normed vector space is a metric space because a norm defines a metric distance

$$d(v, w) = |v - w|.$$

To see this, it is enough to check that it actually satisfies the triangle inequality, because the remaining metric properties follow directly from the norm axioms. Namely,

$$d(v, w) = |v - w| = |v - u + u - w| \leq |v - u| + |u - w| = d(v, u) + d(u, w)$$

holds for any $v, u, w \in V$ due to property 1 of the norm.

Remark: The converse is not true. There are metric distances that do not be obtained from a norm. For instance, given a norm $|\cdot|_a$, the expression

$$d(x, y) = \frac{|x - y|_a}{1 + |x - y|_a}$$

defines a metric distance over $\mathbb{R}^n$ that cannot be written as $d(x, y) = |x - y|_b$ for any norm $|\cdot|_b$.

III. BANACH SPACE

A complete normed vector space is called a *Banach space*. In other words, as we already saw, a normed vector space is a metric space. If additionally, such metric space is complete, meaning that all Cauchy sequences in it converge to vectors in it, then it is called a Banach space.

Remark: Why we need Banach spaces? Why are normed spaces not enough? These questions will be answered in Part II but, essentially, because we want infinite series to be convergent within the space itself.

IV. INNER PRODUCT SPACE

Given a vector space V over $\mathbb{C}$, an *hermitian inner product* is a function $V \times V \ni (v, w) \rightarrow v \cdot w \in \mathbb{C}$ satisfying, for each $v, w, u \in V$ and $\alpha, \beta \in \mathbb{C}$, the following properties:

1. linearity of the second argument, $v \cdot (\alpha w + \beta u) = \alpha(v \cdot w) + \beta(v \cdot u)$,
2. conjugate symmetry, $v \cdot w = (w \cdot v)^*$, and
3. positive definiteness, $v \cdot v \geq 0$ and $v \cdot v = 0$ only if $v = 0$.

A vector space equipped with an (hermitian) inner product is called an inner product space.

Remark: We adopt the physics convention where the inner product is linear in the second argument and antilinear in the first.

A. Induced norm

The mapping $V \ni v \rightarrow \sqrt{v \cdot v}$ is a norm called the *induced norm*. Hence, an inner product space is a normed space.

B. Cauchy-Schwartz inequality

An hermitian inner product satisfies the so called Cauchy-Schwartz inequality

$$|v \cdot w| \leq |v||w|$$

which holds for all $v, w \in V$.

Exercise: Prove the Cauchy-Schwartz inequality.

V. HILBERT SPACE

A Hilbert space is a complete hermitian inner product space. In other words, in a Hilbert space H , all Cauchy sequences converge to a vector in H , where the induced norm is used to determine convergence.

Remark: Why we need Hilbert spaces? Why are inner product spaces not enough? As with Banach spaces, these questions will be answered in Part II. These questions will be answered in Part II but, essentially, because we want linear combinations with infinite terms to be convergent within the space itself.

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